

1930, Blumlein

Stereo mic techniques

Panning laws (sine law), XY and MS stereo techniques

1973, Gerzon

First-Order Ambisonics (FOA)

Same principle as MS but with 3 mics instead of 1: WXYZ(B-format) Later introduces directivity vector $\mathbf{r}_E$

1999, Jot

Binaural Ambisonics through virtual speakers

Decode Ambisonics to a virtual ideal loudspeaker layout then convolve each virtual loudspeaker with the corresponding HRIR

2010, Avni

HRTFs in the SH domain

Allows interpolation through the inverse spatial Fourier transform

2013, Bernschütz

Adaptive Low-Frequency Extension (LFE)

Replace the LF band of measured HRIRs with a synthetic spherical model

2018, Zaunschirm

Time Alignment + diffuseness Constraint (TAC)

Adapt HRIRs for low-order rendering: remove the linear phase ITD (onsets) in the HF band and match original/adapted HRIR covariances via L2

2018, Armstrong

Bi-RADIAL Ambisonics

Binaural rendering with two sets of virtual loudspeakers, one centred at each ear; gains up to two Ambisonic orders

2021, Ben-Hur

Ear time-alignment for bilateral Ambisonics

Like Bi-radial but with direct rendering: TAC, time-aligning the HRIR with respect to the ear. Two soundfields

1964, Wendt et al.

Perceptual properties of auditory events

Elaborates on ITD and ILD

1997, Pulkki

Vector-Based Amplitude Panning (VBAP)

Decompose space into speaker triplets and pan inside one

2000, Daniel

High-Order Ambisonics (HOA) on regular layouts. HF phase issues in basic decoding. Mitigation: weight the SHs so all secondary lobes of the virtual mics are attenuated ($\max \mathbf{r}_E$) suppressed (in-phase).

2012, Zotter

All-Round Ambisonics Decoding (AllRAD)

Physics-based HOA decoder: decode to the virtual loudspeaker signals of an ideal layout, then use them as VBAP sources

2017, Ben-Hur

SpecEQ

$\max \mathbf{r}_E$ causes a HF roll-off, compensated by a global, direction-independent (diffuse) and order-dependent inverse filter

2018, Schörkhuber

Magnitude Least Squares (MagLS)

Alternative to TAC, another direct binaural renderer. In HF: optimise L2 on magnitudes, disregarding phase

2019, Hold

Tapering

Same principle as $\max \mathbf{r}_E$: gradually attenuate the virtual-mic side lobes (reducing angle-dependent artefacts) by windowing the SHs

2021, Engel

Bilateral Magnitude Least-Squares (BiMagLS)

HRTFs are ear (bilateral) time-aligned too. LF from an L2 fit of the higher-order version; HF disregards phase as in MagLS